

A Project Report submitted in partial fulfillment of the requirements for the award
of degree of Bachelor of Technology in

**Computer Science and Engineering (Artificial Intelligence and Machine
Learning)**

Submitted by:

Under the Guidance of

Department of Computer Science and Engineering (AI & ML)

JSS Academy of Technical Education

Noida, Uttar Pradesh

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RE-DACT: An AI-Driven Platform for Automated Redaction of Sensitive Information in Documents

**Submitted to the Department of Computer Science and
Engineering-AIML**

**in Partial Fulfillment of the Requirements
for the Degree of**

Bachelor of Technology

in

Computer Science and Engineering-AIML

by

Name (Roll No.)

Name (Roll No.)

Name (Roll No.)

Name (Roll No.)

Under the Supervision of

Supervisor Name

Department of Computer Science and Engineering-AIML

JSS Academy of Technical Education ,Noida



JSS ACADEMY OF TECHNICAL EDUCATION,NOIDA

DR. APJ ABDUL KALAM TECHNICAL UNIVERSITY

UTTAR PRADESH, LUCKNOW

(Formerly Uttar Pradesh Technical University, Lucknow)

MAY, 2026

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PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

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C410.2: Communicate effectively in verbal and written form

C410.3: Apply appropriate computing, and engineering skills for obtaining solution to the formulated problem within a stipulated time.

C410.4: Work effectively as a part of team in multi-disciplinary areas.

C410.5: Consolidate the final outcome in the form of a publication.

CO-PO-PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
C410.1	3	3	3	3	2	3	3	3	3	3	2	3	3	3
C410.2	2	2	2	2	2	2	2	2	2	3	2	3	2	3
C410.3	3	3	3	3	3	3	2	3	3	3	3	3	3	3
C410.4	3	3	3	3	2	3	2	3	3	3	3	3	3	3
C410.5	3	3	3	3	3	3	3	3	3	3	3	3	3	3
C410	2.80	2.80	2.80	2.80	2.40	2.80	2.40	2.80	2.80	3.00	2.60	3.00	2.80	3.00

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Name : Student Name1

Roll. No. :

(Candidate Signature)

Name : Student Name2

Roll. No. :

(Candidate Signature)

Name : Student Name3

Roll. No. :

(Candidate Signature)

Name : Student Name4

Roll. No. :

(Candidate Signature)

CERTIFICATE

This is to certify that Project Report entitled

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which is submitted by in partial fulfillment of the requirement for the award of degree B. Tech. in Department of Computer Science and Engineering-AIML of Dr. APJ Abdul Kalam Technical University, Uttar Pradesh, Lucknow is a record of the candidates own work carried out by him/her under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Signature

Name of Supervisor

Designation

Address

Date:

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Name : Student Name1

Roll. No. :

(Candidate Signature)

Name : Student Name3

Roll. No. :

(Candidate Signature)

Name : Student Name1

Roll. No. :

(Candidate Signature)

Name : Student Name4

Roll. No. :

(Candidate Signature)

RE-DACT: An AI-Driven Platform for Automated Redaction of Sensitive Information in Documents

ABSTRACT ABSTRACT

Title: RE-DACT: An AI-Driven Platform for Automated Redaction of Sensitive Information in Documents

The proliferation of digital documents containing sensitive personally identifiable information (PII) has created an unprecedented challenge for organizations across healthcare, finance, legal, and government sectors. Manual redaction methods prove inadequate when processing the vast volumes of documents generated daily, leading to potential data breaches and regulatory non-compliance. This project presents RE-DACT, an intelligent web-based platform designed to automatically detect and mask sensitive Indian identity documents including Aadhaar UIDs, PAN cards, and payment card numbers from PDF and image files.

The core innovation of RE-DACT lies in its multi-layered approach to PII detection. The system combines Optical Character Recognition (OCR) using Tesseract for text extraction with mathematical checksum validation algorithms to achieve near-zero false positive rates. Specifically, the Verhoeff algorithm validates Aadhaar 12-digit UIDs while the Luhn algorithm verifies payment card numbers. For PAN cards, structural pattern matching with entity-type validation ensures accurate detection without false positives from similar-looking alphanumeric strings.

A significant technical contribution of this work is the multi-orientation OCR pipeline that tests eight preprocessing combinations comprising four rotation angles and two blur states. This approach addresses the common challenge of arbitrarily oriented documents in real-world scenarios. The pipeline employs an early-exit optimization strategy, terminating processing upon first successful detection, which enables sub-300ms processing for correctly oriented documents while maintaining robust handling of rotated inputs.

The platform implements three configurable masking levels to accommodate diverse compliance requirements. The standard level masks all but the last four characters, partial masking preserves the first four digits, and full masking completely obscures the detected information. Additionally, the system incorporates a tamper-evident audit logging mechanism using SHA-256 hash chains in JSON-lines format, providing cryptographically verifiable compliance records suitable for regulatory audits and forensic analysis.

Empirical evaluation on real Indian identity documents demonstrates consistent performance with processing times ranging from 274 to 1582 milliseconds depending on document complexity and orientation. The Aadhaar detection module achieved a 100% true positive rate on documents with clear, readable text. The platform is built using FastAPI for the REST API layer, with a vanilla HTML/CSS/JavaScript frontend served as static files, ensuring zero-data-retention architecture where processed documents exist only in transient memory during processing.

This report provides comprehensive documentation of the system architecture, detection algorithms, implementation details, and performance evaluation. Future work includes integration of Named Entity Recognition (NER) using transformer-based models like BERT for detecting unstructured PII such as names, addresses, and phone numbers, extending the pipeline to support video redaction, adding multi-language OCR support for Indian scripts, and containerizing the application for enterprise deployment.

Keywords: Redaction, PII, OCR, Verhoeff Algorithm, Luhn Algorithm, Tesseract, Privacy, Compliance, Audit Logging, FastAPI

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CHAPTER 1

INTRODUCTION

1.1 Problem Statement

The digital transformation of modern society has resulted in an exponential increase in the volume of documents containing sensitive personal information. Every day, organizations across healthcare, finance, legal, government, and educational sectors generate and process millions of documents containing personally identifiable information (PII) such as Aadhaar numbers, PAN cards, credit card details, addresses, and phone numbers. This widespread presence of sensitive data has created significant privacy and security challenges that traditional manual redaction methods struggle to address effectively.

The problem of sensitive information exposure in documents manifests in multiple critical ways. First, manual redaction is inherently time-consuming and labor-intensive. Processing a single document manually requires personnel to carefully examine each page, identify sensitive information, and manually black out or mask the identified data. When organizations process thousands of documents daily, this approach becomes completely unscalable. Second, human error introduces significant risk even when personnel attempt manual redaction. Studies have shown that manual redaction achieves accuracy rates between 85-95% under optimal conditions, meaning that 5-15% of sensitive information remains exposed or critical information is improperly removed. Third, the legal and regulatory landscape has become increasingly stringent, with penalties for data exposure reaching into billions of dollars. The European Union's General Data Protection Regulation (GDPR) has imposed fines exceeding 5.88 billion euros since its implementation, including a landmark 1.2 billion euro penalty against Meta in 2023. In India, the Digital Personal Data Protection Act 2023 establishes

comprehensive requirements for handling personal data, with specific provisions for penalty imposition for non-compliance.

The specific challenge this project addresses is the automated detection and redaction of Indian identity documents from digital files. India operates one of the world’s largest identity systems through the Aadhaar program, which has enrolled over 1.3 billion residents. The Permanent Account Number (PAN) system managed by the Income Tax Department serves as another critical identifier for financial transactions. Payment card usage continues to grow exponentially with the digital payments ecosystem. The combination of widespread Aadhaar and PAN card usage in everyday transactions, combined with the increasing digitization of records across government and private sectors, creates a substantial attack surface for identity theft and fraud when documents are improperly handled or disposed of.

Existing redaction solutions in the market present several limitations. Commercial products like Adobe Acrobat offer basic redaction capabilities but lack specialized detection for Indian identity documents and do not implement mathematical checksum validation for detected numbers. Enterprise solutions often require expensive licensing and cloud processing, raising data privacy concerns as sensitive documents must be transmitted to external servers. Open-source alternatives like Microsoft Presidio provide pattern-based detection but primarily target Western document formats and lack the specific detection logic for Indian PII types. Most significantly, no existing solution combines the three critical features this project implements: mathematical checksum validation for detection accuracy, multi-orientation document handling for robust OCR, and tamper-evident audit logging for compliance verification.

1.2 Motivation

The motivation for this project stems from a confluence of practical requirements, technological opportunities, and the specific context of the Indian digital ecosystem.

From a practical standpoint, the handling of identity documents in Indian organizations presents unique challenges. Government offices, financial institutions, healthcare providers, educational institutions, and private businesses regularly process documents containing Aadhaar numbers, PAN card details, and payment information. The lack of

specialized automated tools forces these organizations to rely on manual processes that are neither scalable nor reliable. During the COVID-19 pandemic, the need for vaccination certificates and identity verification dramatically increased document processing volumes, exposing the inadequacy of existing manual redaction capabilities. Similarly, the implementation of direct benefit transfer schemes, where government subsidies are transferred directly to Aadhaar-linked bank accounts, necessitates careful handling of identity documents throughout the process chain.

The technological opportunity arises from significant advances in optical character recognition and checksum validation algorithms that can be combined into a robust detection pipeline. Tesseract, the open-source OCR engine, has achieved maturity with support for LSTM-based neural network recognition, achieving accuracy rates sufficient for document processing. The mathematical foundations of error-detecting codes, specifically the Verhoeff algorithm for Aadhaar validation and the Luhn algorithm for payment card validation, provide deterministic verification that eliminates false positives from random number sequences. By combining these established algorithms with modern web technologies, the project creates a solution that is simultaneously more accurate and more accessible than existing commercial offerings.

The open-source nature of the solution addresses a critical gap in the current market. Organizations requiring PII redaction capabilities often face a difficult choice between expensive commercial products with uncertain data handling practices and basic open-source tools lacking enterprise features. This project demonstrates that it is possible to build a comprehensive solution using freely available technologies without compromising on features or accuracy. The zero-data-retention architecture ensures that sensitive documents never persist beyond the processing operation, addressing the primary concern organizations have about adopting cloud-based or third-party redaction services.

Furthermore, the project addresses an often-overlooked aspect of document processing: orientation handling. Real-world documents arrive in various orientations due to scanning artifacts, mobile photography, or physical document placement during digitization. Standard OCR systems assume correct orientation and fail when documents are rotated, leading to missed detections and incomplete redaction. The multi-orientation scanning approach implemented in this project systematically tests

multiple orientations, ensuring robust detection regardless of document positioning.

The educational value of the project also deserves mention. Building a complete document processing system provides hands-on experience with multiple technologies including OCR, cryptographic hash chains, web services, and algorithm implementation. The project structure serves as a reference implementation for similar systems and provides a foundation for further research and development in the field.

1.3 Project Objectives

The primary objective of this project is to design and develop a web-based platform that automatically detects and redacts sensitive personally identifiable information from PDF and image documents, with specific focus on Indian identity documents including Aadhaar UIDs, PAN cards, and payment card numbers.

The project encompasses the following specific objectives:

The first objective involves building a robust PII detection system that combines multiple detection strategies to achieve high accuracy. This includes implementing the Verhoeff checksum algorithm for validating Aadhaar 12-digit numbers, the Luhn algorithm for validating payment card numbers of 13-19 digits, and structural pattern matching with entity-type validation for PAN card numbers. The detection system must operate on text extracted through OCR, converting the extracted characters into a searchable format while preserving positional information for subsequent masking operations.

The second objective requires developing a multi-orientation OCR pipeline that handles documents in various rotational states. The pipeline must test multiple pre-processing combinations including four rotation angles (0°, 90°, 180°, 270°) and two image preprocessing states (original and Gaussian-blurred). The system must implement early-exit optimization to ensure reasonable processing times when documents are correctly oriented, while maintaining comprehensive scanning capability for rotated inputs.

The third objective concerns implementing configurable masking levels that cater to different compliance requirements. The system must provide three distinct masking strategies: standard masking that preserves the last four characters for reference,

partial masking that preserves the first four characters, and full masking that completely obscures all detected information. Each masking strategy must be applied at the character level within detected PII regions.

The fourth objective focuses on creating a complete web application with REST API endpoints for programmatic access and an intuitive user interface for manual operations. The web interface must support drag-and-drop file upload, real-time processing status, and result visualization including detection statistics and preview of redacted documents.

The fifth objective involves implementing tamper-evident audit logging that maintains verifiable records of all processing operations. The audit system must use cryptographic hash chains to ensure that historical records cannot be modified without detection, providing compliance evidence suitable for regulatory audits and forensic analysis.

The sixth objective requires achieving acceptable processing performance with the goal of sub-300ms processing time for single-page images with correctly oriented text. The system must handle both PDF documents (with multi-page support) and common image formats (JPG, JPEG, PNG) while maintaining memory efficiency through appropriate cleanup and timeout mechanisms.

1.4 Scope of Project

The scope of this project defines the boundaries of the developed system, clarifying what the project addresses and what falls outside its intended purpose.

The project encompasses the following in-scope elements. First, the detection and redaction of three specific PII types from Indian identity documents: Aadhaar UIDs validated through Verhoeff checksum, payment card numbers validated through Luhn algorithm, and PAN card numbers validated through structural pattern matching. Second, the system processes two document categories: PDF files (with support for multiple pages converted to images) and image files in JPG, JPEG, and PNG formats. Third, the platform implements three configurable redaction levels as described previously. Fourth, the system provides a web-based user interface with drag-and-drop upload, redaction level selection, and result visualization. Fifth, the platform exposes

REST API endpoints for programmatic access by other applications. Sixth, the system implements tamper-evident audit logging with SHA-256 hash chain verification. Seventh, the system maintains a zero-data-retention architecture where processed documents exist only in transient memory during the processing operation.

The project explicitly excludes certain elements from its scope. Video redaction capability remains outside the current implementation, though this is identified as a direction for future work. Multi-language OCR support focusing on English-language text extraction is the current limitation, with future plans to extend support for Devanagari, Tamil, and other Indian scripts. The system does not include integration with external storage systems or cloud platforms. Real-time document processing for streaming video content is not supported in the current version. The system does not provide batch processing capabilities for handling large volumes of documents in scheduled operations.

The assumptions underlying the project scope include that input documents will contain readable printed or typed text rather than handwritten content. The OCR accuracy depends on image quality, and the system is optimized for scanned documents or photographs of physical documents rather than screenshots of digital documents. The project assumes a standalone deployment model rather than distributed or clustered processing architecture. The implementation focuses on single-user request handling rather than concurrent bulk processing scenarios.

The constraints affecting the project include processing performance dependent on the client machine’s specifications, particularly for PDF conversion and OCR operations. The NER-based detection feature requires the transformers and torch libraries, with graceful fallback to rule-based detection when these are unavailable. The system design prioritizes accuracy over speed, accepting the computational cost of multiple orientation testing to ensure comprehensive detection.

1.5 Organization of the Report

This report is organized into five chapters and supporting sections to provide comprehensive documentation of the project from conceptualization through implementation and evaluation.

Chapter 1, the Introduction, establishes the problem context, motivation, objectives, and scope. This chapter explains why automated PII redaction is necessary, what specific challenges the project addresses, and what boundaries define the system’s functionality.

Chapter 2, Literature Review, presents the background and related work in the field of document redaction and PII detection. This chapter examines traditional approaches, AI-based methods, existing tools and their limitations, and identifies the specific research gap that this project fills. The chapter also covers relevant privacy regulations and their implications for document handling.

Chapter 3, System Design and Methodology, describes the architectural design and algorithmic approaches used in the implementation. This chapter provides detailed explanations of the Verhoeff and Luhn validation algorithms, the multi-orientation OCR pipeline, masking strategies, overlap deduplication logic, and the tamper-evident audit logging mechanism. Technical details sufficient for understanding and replicating the system are provided throughout.

Chapter 4, Implementation and Results, presents the practical implementation details and experimental evaluation of the system. This chapter covers the software and hardware requirements, API design, frontend implementation, test case execution, processing time benchmarks, and performance comparison with existing solutions. Screenshots and visualizations of the system interface are included where appropriate.

Chapter 5, Conclusion and Future Scope, summarizes the achievements of the project and outlines directions for future enhancement. The chapter reflects on the extent to which project objectives were met, identifies areas for improvement, and proposes extensions including NER-based detection, video redaction support, multi-language OCR, and containerized deployment.

Following the main chapters, the References section provides complete bibliographic information for all sources cited throughout the report. The Appendices contain supplementary material including lists of published work and plagiarism reports as required by the academic institution.

This organizational structure ensures that readers can progressively build understanding from foundational concepts through system design to practical implementation, with sufficient technical depth to support both academic evaluation and practical

replication of the system.

CHAPTER 2

LITERATURE REVIEW (NON-WEB BASED PROJECTS/SOFTWARE REQUIREMENT SPECIFICATION(WEB BASED PROJECTS))

2.1 Background and Terminology

The domain of document redaction and personally identifiable information (PII) protection has evolved significantly over the past decades, driven by increasing digital document volumes and tightening privacy regulations. Understanding the fundamental concepts and terminology provides essential foundation for appreciating the technical approaches examined in this chapter.

Personally Identifiable Information refers to any data that can be used to identify, contact, or locate a single person, or to identify an individual in context. The concept extends beyond simple identifiers like names or identification numbers to include information that can be combined to reveal an individual's identity. The Information Technology Act of India, through its various amendments and rules, defines sensitive personal data to include biometric information, purpose of processing, and data that requires higher protection levels. The Digital Personal Data Protection Act of 2023 establishes a comprehensive framework for processing digital personal data, imposing specific obligations on data fiduciaries regarding collection, storage, and protection of personal information.

The context of Indian identity documents presents unique characteristics that differentiate it from Western identification systems. The Aadhaar system, established in 2009, provides a 12-digit unique identification number linked to demographic and biometric information of residents. The Aadhaar number follows specific structural

patterns and incorporates a Verhoeff checksum digit for validation, making algorithmic verification possible beyond simple pattern matching. The Permanent Account Number (PAN) is a ten-character alphanumeric identifier issued by the Income Tax Department for financial transaction tracking. PAN card numbers follow a specific format where the first five characters are uppercase letters, the next four are digits, and the final character is a letter representing the entity type. Payment cards issued by banks and financial institutions follow international standards with 13-19 digit numbers validated through the Luhn algorithm.

Redaction, in the context of document processing, refers to the process of removing, obscuring, or masking sensitive information from documents before sharing or publishing. Traditional manual redaction involves physically marking or cutting portions of physical documents, while digital redaction uses software tools to overlay black boxes or remove text from electronic files. The effectiveness of redaction is measured by two key metrics: precision (correctly identifying and masking actual sensitive information) and recall (not missing any sensitive information that should be masked). A failed redaction, where sensitive information remains visible despite attempted masking, can lead to serious consequences including identity theft, financial fraud, and regulatory penalties.

2.2 Traditional Redaction Approaches

Manual redaction represents the most traditional approach to document sanitization, involving human reviewers examining each document to identify and mask sensitive information. This approach, while straightforward in concept, presents substantial practical challenges. The average human attention span and processing speed limit the volume of documents that can be reviewed manually. Typical processing rates range from 5-15 pages per hour depending on document complexity, making large-scale processing operations prohibitively time-consuming and expensive. The error rate in manual redaction presents another significant concern. Cognitive psychology research demonstrates that repetitive tasks lead to attention lapses, and simple errors like missing a single digit in a long number can render the entire redaction ineffective.

Rule-based pattern matching using regular expressions provides the first layer of

automated redaction capability. This approach uses predefined patterns to identify potential sensitive information in text. For example, a regular expression matching exactly twelve consecutive digits identifies candidate Aadhaar numbers, while patterns matching the format ‘AAAAA9999A’ identify candidate PAN numbers. Regular expression systems offer the advantages of transparency (the matching logic is explicit and auditable), speed (modern regex engines process text efficiently), and simplicity (patterns can be written and maintained without machine learning expertise).

However, pure regex-based approaches suffer from fundamental limitations. They cannot distinguish between valid and invalid identification numbers that happen to match the structural pattern. A twelve-digit sequence like ‘123456789012’ matches the pattern for Aadhaar but is not a valid number because it fails checksum validation. The result is false positives that lead to over-redaction, obscuring legitimate non-sensitive numbers. Conversely, variations in formatting break pattern matching entirely. Aadhaar numbers may be written with spaces as ‘1234 5678 9012’ or with dashes as ‘1234-5678-9012’, and simple patterns fail to capture these variations. Furthermore, regex patterns cannot understand context. A twelve-digit number appearing in a page number, a document ID, or a table of statistics would trigger false redaction, and determining whether the number represents an actual Aadhaar requires understanding the document semantics.

Traditional OCR systems like early versions of Tesseract relied purely on pattern recognition without validation, producing text output that required downstream processing to identify sensitive information. The accuracy of OCR itself varies dramatically based on image quality, font characteristics, and scanning conditions. Degraded documents with noise, skew, or poor contrast produce OCR outputs with character recognition errors that make downstream pattern matching unreliable. The assumption in many traditional systems that scanned documents arrive in standard orientation fails in practice where documents may be rotated, skewed, or photographed at angles.

2.3 AI-Based Approaches for PII Detection

The emergence of machine learning and deep learning approaches has transformed the landscape of document analysis and PII detection. These technologies offer the poten-

tial for context-aware detection, adaptability to new document types, and handling of unstructured information that rule-based systems struggle with.

Optical Character Recognition using modern engines achieves significantly higher accuracy than earlier systems through the application of neural networks. Tesseract version 5, the current stable release used in this project, implements Long Short-Term Memory (LSTM) neural networks for character recognition. The LSTM architecture excels at sequential data processing, making it well-suited for reading text where the identity of each character depends on both the character’s visual features and the surrounding context. The Tesseract engine offers multiple OCR Engine Modes (OEM) with mode 3 (default and most comprehensive) utilizing a combination of LSTM neural networks and legacy recognition engines. Page Segmentation Modes (PSM) allow configuration for different document layouts, with PSM 11 (sparse text) optimized for documents containing scattered numeric identifiers like identity documents.

Named Entity Recognition (NER) represents a significant advancement beyond rule-based pattern matching by enabling context-aware identification of information types. Modern NER systems use transformer architectures like BERT (Bidirectional Encoder Representations from Transformers) to understand word context from large amounts of training data. The `dslim/bert-base-NER` model, used in this project for detecting names and addresses, identifies PERSON and LOCATION entities with confidence scores that can be thresholded for accuracy control. NER systems distinguish between similar-looking entities that rules cannot differentiate. For instance, the word "John" might be a person’s name (requiring redaction) or part of a company name "John’s Bakery" (which may or may not require redaction depending on context), and NER models trained on diverse text can make this distinction.

Deep learning object detection extends redaction capabilities beyond text to visual elements in images. Systems like YOLO (You Only Look Once) can identify faces, license plates, and other visual PII in photographs and video frames. This capability is essential for complete redaction of documents containing both printed text and visual identifiers. However, these systems require substantial computational resources and training data, making them complex to deploy in resource-constrained environments.

Large Language Models (LLMs) represent the cutting edge of natural language processing, offering sophisticated contextual understanding that can theoretically handle

complex redaction scenarios. Research has explored using LLMs for PII detection and redaction, with findings indicating that fine-tuned models can achieve high accuracy on structured PII types. However, LLMs present challenges including computational cost (both inference time and hardware requirements), data privacy concerns (sending documents to external API services), and consistency issues (the same prompt may produce different outputs). The benchmark study by Garza et al. on the PRvL dataset demonstrates that while LLMs offer powerful generalization capabilities, their performance varies significantly across architectures and adaptation methods, revealing complex trade-offs between accuracy, latency, and PII leakage risks.

The combination of multiple detection approaches offers the most robust solution. Rule-based checksum validation provides mathematical certainty for structured identifiers like Aadhaar and payment card numbers. Pattern matching handles formats like PAN numbers that lack checksums. NER adds capability for unstructured PII like names and addresses. This layered approach, implemented in RE-DACT, achieves detection accuracy that exceeds any single method alone.

2.4 Related Work in Document Redaction

The research community has produced significant work addressing various aspects of automated redaction, establishing theoretical foundations and demonstrating practical implementations.

Sánchez and Batet introduced the "C-sanitized" model, a framework grounded in information theory that provides a priori privacy guarantees for document redaction. Their approach moves beyond simple pattern matching to consider semantic inferences that could lead to data disclosure, quantifying disclosure risk using metrics like Information Content (IC) and Point-wise Mutual Information (PMI). While pioneering, this model remained primarily theoretical, highlighting the complexity of achieving truly semantic-aware redaction in practice.

Lee and Woods explored automated redaction in the context of digital heritage and archives, examining digital forensics tools for redacting private data from complex digital artifacts like disk images. Their work on the BitCurator project demonstrated the necessity of specialized workflows for handling born-digital collections, using tools

like bulk extractor to identify sensitive data patterns. This research emphasized that effective redaction requires different strategies for different data contexts, from simple documents to entire file systems.

Orekondy et al. addressed the redaction of private information in images by proposing a "redaction by segmentation" paradigm. This approach identifies and obfuscates specific pixel regions corresponding to privacy attributes including faces, fingerprints, and license plates. Their work emphasized the critical trade-off between privacy and utility, showing that targeted region redaction can preserve image intelligibility while protecting sensitive information. This highlighted the need for integrated computer vision capabilities in comprehensive redaction platforms.

In the specific context of document processing, Thetbanthad et al. demonstrated a two-stage pipeline using OCR and an LLM to redact PII from drug labels in compliance with Thailand's Personal Data Protection Act (PDPA). Their work illustrated how AI can be tailored to specific regulatory and linguistic contexts, combining Tesseract OCR for text extraction and LLMs for classification. This approach directly influenced the architecture of RE-DACT, which similarly combines OCR with structured detection rather than relying solely on either method.

The controlled experiment by Peng et al. comparing manual redaction with AI-assisted tools found that AI technologies like iDox.ai Redact significantly outperform manual methods in both accuracy and speed. This provided empirical evidence for the revolutionary impact of AI in data protection workflows, though their study focused primarily on commercial tools rather than open-source implementations.

Existing commercial tools provide instructive comparisons for system design. Adobe Acrobat Pro offers redaction capabilities but lacks specialized Indian PII detection and checksum validation. Microsoft Presidio provides open-source PII detection with pattern-based and NER-based detection, but requires customization for Indian document types. iDox.ai offers commercial redaction with some API-based access but lacks transparency about detection methods and does not specifically target Indian identity formats. No existing solution combines the three key features of this project: mathematical checksum validation, multi-orientation robustness, and cryptographic audit logging.

2.5 Algorithms for PII Validation

The mathematical foundations underlying PII validation in this project deserve detailed examination, as these algorithms provide the deterministic accuracy that distinguishes the system from pattern-matching-only approaches.

The Verhoeff algorithm, developed by Jacobus Verhoeff in 1969, provides error detection for identification numbers through a sophisticated mathematical construction. Unlike simpler checksum algorithms like Mod 10, the Verhoeff algorithm catches all single-digit errors and all adjacent transpositions, making it significantly more robust for validation purposes. The algorithm operates using two tables: a multiplication table (a 10E10 Cayley table of the dihedral group D5) and a permutation table (an 8E10 matrix). For a 12-digit number, the algorithm processes digits from right to left, using the position index to select the appropriate permutation value for each digit, then combining through the group operation. The final checksum of zero indicates a valid number. The probability that a random 12-digit number passes the Verhoeff check is approximately 1 in 10, providing strong false positive rejection.

The Luhn algorithm, invented by Hans Peter Luhn in 1954 and patented in 1960, serves as the international standard for validating credit card numbers and other identification numbers. The algorithm applies a modulus-10 check where every second digit from the right is doubled, and if the result exceeds 9, the digits are summed (equivalent to subtracting 9). The sum of all digits must be divisible by 10 for the number to be valid. While simpler than Verhoeff, the Luhn algorithm effectively catches single-digit errors and transposition errors involving adjacent digits, making it suitable for payment card validation where the volume of potential numbers is extremely high.

PAN card validation differs fundamentally from Aadhaar and payment card validation because PAN numbers do not incorporate a checksum. Instead, validation relies entirely on structural pattern matching. The PAN format '[A-Z]5[0-9]4[A-Z]' ensures a specific structure, but the fourth character carries additional meaning as an entity type indicator. Valid entity type characters include A (Association of Persons), B (Body of Individuals), C (Company), F (Firm), G (Government), H (Hindu Undivided Family), J (Artificial Juridical Person), L (Local Authority), P (Individual), and T (Trust). This entity type validation adds a layer of semantic checking beyond simple pattern

matching, reducing false positives from random alphanumeric sequences.

These validation algorithms provide mathematical certainty that cannot be achieved through pattern matching alone. When combined with OCR text extraction, they create a detection pipeline with near-zero false positive rates on well-structured PII, addressing the primary weakness of rule-based systems.

2.6 Research Gap Analysis

The literature review reveals specific gaps in existing approaches that this project addresses through its integrated design and implementation.

The first identified gap involves the combination of OCR with mathematical validation. Most existing systems use either OCR with pattern matching (susceptible to false positives) or structured validation without context (unable to handle OCR errors and format variations). RE-DACT bridges this gap by extracting text through OCR, identifying candidate sequences through pattern matching, then validating candidates through mathematical algorithms before reporting detections. This two-stage approach achieves detection accuracy that exceeds either method alone.

The second gap concerns multi-orientation document handling. Existing solutions typically assume correct document orientation or require manual pre-processing to correct rotation. The multi-orientation OCR pipeline in RE-DACT systematically tests eight preprocessing combinations, ensuring detection regardless of document positioning while the early-exit optimization maintains reasonable processing times for correctly oriented documents.

The third gap relates to audit trail capabilities. While commercial products may offer logging, the specific requirement for tamper-evident records using cryptographic hash chains is largely unaddressed in existing open-source solutions. The SHA-256 hash chain implementation in RE-DACT provides verifiable evidence suitable for compliance audits, something not available in comparable tools.

The fourth gap involves support for Indian identity documents specifically. Most existing tools target Western document formats (SSN, credit cards, etc.) and require significant customization for Indian PII types. RE-DACT implements explicit detection logic for Aadhaar, PAN, and Indian payment cards, providing immediate utility for

organizations processing Indian documents.

The fifth gap concerns accessibility. Commercial redaction tools require expensive licensing, while academic tools often lack the polish for production use. RE-DACT provides a complete, usable solution as open-source software, enabling adoption by organizations with limited budgets while allowing modification and extension as needed.

These gaps, summarized in Table 2.1, define the specific contribution of this project to the field of automated document redaction.

Table 2.1: Summary of Literature Gaps and RE-DACT’s Contributions

Gap Area	Existing Limitation	RE-DACT Solution
OCR + Validation	Pattern matching without mathematical validation leads to false positives	Two-stage approach: OCR extracts text, then Verhoeff/Luhn validate candidates
Orientation Handling	Assumes correct document orientation, fails on rotated inputs	Multi-orientation pipeline tests 8 pre-processing combinations with early-exit
Audit Trail	Limited logging capability, no tamper evidence	SHA-256 hash chain provides cryptographically verifiable records
Indian PII Support	Tools target Western formats (SSN, credit cards)	Explicit detection for Aadhaar, PAN, Indian payment cards
Accessibility	Expensive commercial tools or academic prototypes lacking polish	Complete, usable open-source solution with web interface

2.7 Summary

This chapter has examined the landscape of document redaction and PII detection through analysis of traditional approaches, AI-based methods, and existing systems. The fundamental limitations of manual redaction (unscalable, error-prone) and simple pattern matching (false positives, context-blind) establish the need for more sophis-

ticated approaches. The emergence of neural network-based OCR, transformer-based NER, and mathematical validation algorithms creates opportunities for systems that achieve high accuracy while remaining accessible.

The research gap analysis identifies five specific areas where existing solutions fall short, all of which RE-DACT addresses through its integrated design. The combination of OCR with checksum validation, multi-orientation robustness, cryptographic audit logging, specialized Indian PII support, and open-source accessibility represents a meaningful contribution to the field.

The next chapter examines the system architecture and algorithmic approaches in detail, providing the technical foundation for understanding the implementation described in subsequent chapters.

CHAPTER 3

SYSTEM DESIGN AND METHODOLOGY

(NON-WEB BASED/SYSTEM DESIGN(WEB-BASED PROJECTS))

3.1 System Architecture Overview

The RE-DACT system implements a modular, multi-stage pipeline architecture that processes uploaded documents through a series of well-defined stages, each handling a specific aspect of the redaction workflow. The architecture balances detection accuracy with processing performance, incorporating optimizations that enable sub-300ms processing for typical documents while maintaining comprehensive scanning capability for challenging inputs.

The system architecture follows a layered approach comprising the presentation layer, processing layer, and data layer. The presentation layer provides the web-based user interface through which users interact with the system, handling file upload, redaction level selection, and result visualization. The processing layer implements the core redaction logic including document conversion, OCR processing, PII detection, and masking operations. The data layer manages temporary file storage, result caching, and audit logging.

At the core of the architecture is a six-stage processing pipeline that transforms an uploaded document into a redacted output. The pipeline stages are: (1) File Ingestion and Type Detection, (2) PDF-to-Image Conversion at 300 DPI, (3) Multi-Orientation OCR with Preprocessing, (4) PII Detection via Checksum Validation, (5) Coordinate Masking and Output Assembly, and (6) Tamper-Evident Audit Logging. Each stage receives input from the previous stage, processes it according to defined logic, and

passes output to the subsequent stage. This sequential design ensures clear data flow and simplifies debugging and testing.

The architectural design emphasizes several key principles. Modularity allows individual components to be updated or replaced without affecting other parts of the system. For example, the PII detection algorithms can be enhanced or new detection types added without modifying the OCR or masking components. The zero-data-retention principle ensures that processed documents exist only in transient memory during request handling, addressing data privacy concerns. The thread-safe implementation enables concurrent request handling, supporting production deployment with multiple simultaneous users.

The technology stack reflects these architectural choices. The backend uses FastAPI as the web framework, providing high performance and automatic API documentation. The redaction engine uses pure Python implementations of detection algorithms, avoiding heavy dependencies that could complicate deployment. Tesseract handles OCR through the pytesseract wrapper, and OpenCV manages image manipulation operations. The PDF processing uses pdf2image for conversion to images and img2pdf for reconstructing output PDFs from processed images.

The frontend implements a single-page application with vanilla HTML, CSS, and JavaScript. This lightweight approach eliminates the need for framework dependencies while providing a responsive, intuitive interface. The dark-mode design aesthetic aligns with modern design preferences while ensuring readability. The interface provides immediate feedback during processing, displaying progress indicators and processing time metrics.

3.2 PII Detection Module Design

The PII detection module forms the analytical core of the RE-DACT system, implementing specialized algorithms that identify sensitive information in extracted text. The module combines multiple detection strategies, each optimized for specific PII types found in Indian identity documents.

The module’s design follows a unified interface pattern where all detection algorithms conform to a common signature, accepting text input and returning a list of

detection dictionaries. Each detection dictionary includes the type of PII identified, the actual value detected, character position information (start and end indices), length of the detection, and confidence score. This standardized format enables the downstream masking component to process detections uniformly regardless of which algorithm produced them.

The detection module supports five distinct PII types: Aadhaar UUIDs validated through Verhoeff checksum, payment card numbers validated through Luhn algorithm, PAN card numbers validated through structural pattern matching, email addresses detected through regex pattern, and phone numbers detected through regex pattern. Each type has specific validation requirements that determine whether a candidate is considered a legitimate PII or a false positive to be rejected.

The detection process begins with regex pattern matching to identify candidate sequences within the extracted text. Regular expressions are crafted to match the structural format of each PII type without attempting validation. For example, the pattern `'12'` identifies any twelve-digit sequence that could be an Aadhaar number, and the pattern `'[A-Z]5[0-9]4[A-Z]'` identifies any ten-character alphanumeric sequence matching PAN format. These candidates are then passed to the validation functions that apply mathematical or structural checks.

The validation step is crucial for eliminating false positives. Without validation, any twelve-digit number would be flagged as potential Aadhaar, leading to excessive false detections. The validation algorithms use mathematical properties of the identification numbers to distinguish genuine identifiers from random numbers that happen to match the format. This mathematical validation is what enables the near-zero false positive rate that distinguishes RE-DACT from pattern-matching-only systems.

Following validation, the module performs overlap deduplication. When multiple detection algorithms identify the same or overlapping character sequences, the system must determine which detection to retain. The deduplication algorithm sorts detections by position, then iterates through them, keeping the first detection and discarding any that overlap with previously retained detections. For overlapping detections, the algorithm retains the one with higher confidence score or, for equal confidence, the longer detection. This ensures each character in the text is attributed to at most one PII entity.

The module also supports optional Named Entity Recognition (NER) for detecting unstructured PII like names and addresses. When the transformers and torch libraries are available, the module uses a BERT-based NER pipeline to identify PERSON and LOCATION entities in the text. These detections are merged with rule-based detections using the same deduplication logic. When NER libraries are unavailable, the module gracefully falls back to rule-based detection only.

The confidence scoring system enables downstream processing to make informed decisions about detections. Rule-based detections with mathematical validation receive a confidence score of 1.0 (maximum certainty), as the mathematical algorithms provide deterministic validation. NER-based detections receive confidence scores from the underlying BERT model, thresholded at 0.85 by default to balance precision and recall. Users can adjust this threshold through API parameters to tune detection behavior for their specific use cases.

3.3 Verhoeff Algorithm for Aadhaar Validation

The Aadhaar UID validation implements the Verhoeff algorithm, a sophisticated checksum scheme that catches all single-digit errors and all adjacent digit transpositions. This algorithm is particularly well-suited for Aadhaar validation because the Indian government designed the number allocation to support Verhoeff validation, making the algorithm applicable to all genuine Aadhaar numbers.

The mathematical foundation of the Verhoeff algorithm relies on the dihedral group D_5 , which has ten elements corresponding to the digits 0-9. The group operation is defined through a 10×10 multiplication table where each cell contains the result of multiplying two elements. Additionally, a permutation table of size 8×10 specifies how each digit should be transformed based on its position in the number. The algorithm iterates through the digits from right to left, using the position index to select the appropriate permutation value for each digit, then combining this with the running checksum through the group operation.

The implementation uses two static tables defined at module level for efficiency. The `VERHOEFF_MULT` table contains the multiplication table for the dihedral group, a 10×10 matrix where each row represents one element's multiplication results with the

digits 0-9. The VERHOEFF_PERM table contains the position-based permutation matrix, an 8E10 matrix where each row specifies how the digits should be permuted at that position. The eight rows correspond to the eight possible positions in a 12-digit number (positions 0-7 repeat for longer numbers).

The `verhoeff_checksum` function takes a number string as input and returns an integer checksum value. The function converts the number string to a tuple of integer digits in reversed order (processing from right to left), initializes the checksum to zero, then iterates through each digit. For each digit at position *i*, the function updates the checksum using the formula: `checksum = VERHOEFF_MULT[checksum][VERHOEFF_PERM[i]`

The `detect_aadhaar` function applies Verhoeff validation to identify valid Aadhaar numbers. It searches the text for twelve-digit sequences using regex, then calls `verhoeff_checksum` on each candidate. Candidates with checksum equal to zero are valid Aadhaar numbers. The function applies an additional filter rejecting numbers ending in 1947, which addresses a specific false positive pattern observed in real documents where common year references match the Aadhaar format but are not actual identification numbers.

Pseudo-code for the Aadhaar detection process follows:

```

1: Input: Image I (via OCR text extraction)
2: Output: List of valid Aadhaar detections
3: Extract text and bounding boxes B from I via Tesseract OCR.
4: Find all 12-digit candidate strings S using regex '12'.
5: for each candidate s in S do
6:   c ← VerhoeffChecksum(s)
7:   if c = 0 AND s mod 10000 ≠ 1947 then
8:     Get coordinates for s from bounding boxes B.
9:     Add detection to results with type "aadhaar", position, and confidence 1.0.
10:  end if
11: end for
12: return Results list

```

The algorithmic complexity of Aadhaar validation is $O(n)$ where *n* is the number of digits (12 for Aadhaar). This makes validation computationally trivial compared

to OCR processing. The deterministic nature of the algorithm means that the same input always produces the same output, enabling consistent detection across multiple processing runs.

3.4 Luhn Algorithm for Payment Card Validation

The payment card validation implements the Luhn algorithm (also known as modulus-10 or mod-10 algorithm), which serves as the international standard for validating credit card numbers, IMEI numbers, and other identification numbers. The algorithm detects common transcription errors including single-digit errors and most adjacent transpositions.

The Luhn algorithm operates by computing a weighted sum of the digits in the number and checking whether this sum is divisible by 10. The weighting alternates for each digit: digits in even positions from the right (including the rightmost check digit at position 0) are used unchanged, while digits in odd positions are doubled. When doubling a digit results in a value greater than 9, the digits of the result are summed (equivalent to subtracting 9). The sum of all weighted digits must be divisible by 10 for the number to be valid.

The `luhn_checksum` function implements this algorithm in Python. The function converts the number string to a list of integer digits, then iterates through the list from the third-from-last digit to the first, doubling every second digit. For doubled digits exceeding 9, the function subtracts 9 to get the correct contribution. Finally, the function returns `True` if the sum modulo 10 equals zero.

The `detect_payment_cards` function applies Luhn validation to identify valid payment card numbers. It searches the text for digit sequences of 13-19 digits (the valid range for payment card numbers), then applies Luhn validation to each candidate. The function maintains a separate detection type `payment_card` to distinguish these from Aadhaar detections.

The payment card detection also implements overlap handling with Aadhaar detections. When a 12-digit valid Aadhaar appears as the prefix of a longer valid payment card number, the system must decide how to handle the overlap. The current implementation prioritizes Aadhaar detection (due to the specific nature of the Indian identity system), meaning that 12-digit sequences validated as Aadhaar are reported

as such rather than as partial payment card numbers.

Pseudo-code for payment card detection:

```
1: Input: Image  $I$  (via OCR text extraction)
2: Output: List of valid payment card detections
3: Extract text and bounding boxes  $B$  from  $I$  via Tesseract OCR.
4: Find all 13-19 digit candidate strings  $S$  using regex '13,19'.
5: for each candidate  $s$  in  $S$  do
6:   if  $LuhnChecksum(s) = 0$  then
7:     Get coordinates for  $s$  from bounding boxes  $B$ .
8:     Add detection to results with type "payment_card", position, and confidence
       1.0.
9:   end if
10: end for
11: return Results list
```

The computational complexity of Luhn validation is $O(n)$ where n is the number of digits (13-19 for payment cards). Like Verhoeff validation, this is computationally trivial compared to text extraction. The algorithm's widespread standardization means it correctly validates all major card schemes including Visa, Mastercard, American Express, and RuPay.

3.5 PAN Card Detection with Entity Validation

PAN card detection differs fundamentally from Aadhaar and payment card validation because PAN numbers do not incorporate a mathematical checksum. Instead, validation relies entirely on structural pattern matching with additional semantic constraints from the entity type field.

The structural formulation defines a valid PAN as a ten-character string matching the pattern: first five characters are uppercase letters (A-Z), next four characters are digits (0-9), and the final character is an uppercase letter. This pattern alone filters out most random alphanumeric sequences but still accepts invalid PANs that happen to match the format.

The entity type validation adds a semantic constraint to structural matching. The

fourth character of a PAN encodes the holder’s entity type according to the following mapping: A represents Association of Persons, B represents Body of Individuals, C represents Company, F represents Firm, G represents Government, H represents Hindu Undivided Family, J represents Artificial Juridical Person, L represents Local Authority, P represents Individual, and T represents Trust. Only PANs where the fourth character falls within this valid set are considered genuine.

The implementation uses compiled regular expressions for efficient pattern matching. The PAN_REGEX pattern ‘[A-Z]5[0-9]4[A-Z]’ matches ten-character alphanumeric sequences matching the PAN structure. The VALID_PAN_4TH_CHARS set contains the valid entity type characters. The detect_pan function applies both the pattern match and the entity type validation, adding detections only when both conditions are satisfied.

Pseudo-code for PAN detection:

```

1: Input: Image  $I$  (via OCR text extraction)
2: Output: List of valid PAN card detections
3: Extract text and bounding boxes  $B$  from  $I$  via Tesseract OCR.
4: Find all 10-character alphanumeric candidate strings  $S$  using regex ‘[A-Z]5[0-9]4[A-Z]’.
5: for each candidate  $s$  in  $S$  do
6:   if  $s[3]$  (4th character) is in VALID_PAN_4TH_CHARS then
7:     Get coordinates for  $s$  from bounding boxes  $B$ .
8:     Add detection to results with type "pan", position, and confidence 1.0.
9:   end if
10: end for
11: return Results list

```

The PAN detection relies on proper OCR output being in uppercase. Tesseract’s configuration includes the ‘-psm 11’ (sparse text) mode optimized for scattered text like that found on identity documents, and the default behavior outputs uppercase for alphabetic characters. However, the OCR may produce lowercase output in some cases, requiring pre-processing to convert text to uppercase before PAN detection.

The algorithmic complexity of PAN detection is $O(n)$ where n is the text length,

dominated by the regex matching operation. The detection does not involve iterative computation like the Verhoeff or Luhn algorithms, making it computationally lightweight.

3.6 Named Entity Recognition for Unstructured PII

Beyond the structured PII types with mathematical validation, RE-DACT optionally incorporates Named Entity Recognition (NER) for detecting unstructured personally identifiable information such as names and addresses. This capability extends the system's detection beyond format-specific identifiers to general PII categories.

The NER implementation uses the `dslim/bert-base-NER` model from Hugging Face Transformers, a BERT-based model fine-tuned for named entity recognition on the CoNLL-2003 dataset. The model identifies PERSON entities (individual names) and LOCATION entities (places, addresses) with associated confidence scores. A threshold parameter (default 0.85) filters out low-confidence detections that may represent false positives.

The implementation handles the fundamental challenge of applying transformer-based NER to OCR output: token length limitations. BERT models have a maximum token limit of 512 tokens, which is sufficient for most single-page documents but requires chunking for longer texts. The `detect_ner_pii` function implements overlapping chunk processing to handle texts exceeding the token limit. The text is divided into chunks of 512 tokens with 50-token overlap between chunks to ensure entities spanning chunk boundaries are captured. Detections from overlapping chunks are merged using a position-based deduplication strategy.

The NER pipeline requires the `transformers` and `torch` libraries, which are optional dependencies. When these libraries are not available, the system gracefully falls back to rule-based detection only, maintaining basic functionality without the additional NER capability. The `is_ner_available` function checks for library availability at startup, and the system reports NER capability status through the health endpoint.

The NER label mapping translates model output to internal PII types: PER (Person) maps to "name" and LOC (Location) maps to "address". These detection types receive lower confidence scores than mathematically validated detections, reflecting

the probabilistic nature of NER compared to deterministic validation. The confidence threshold ensures only high-quality detections are reported, though users can adjust this parameter through the API.

The use case for NER extends the system’s applicability beyond specifically formatted identification numbers to general sensitive information. A document might contain a person’s name, home address, or organization name that should be redacted but does not follow the specific patterns of Aadhaar, PAN, or payment card numbers. The NER capability handles these cases, making the system more comprehensive in its protection capabilities.

The computational cost of NER significantly exceeds rule-based detection. Loading the BERT model requires substantial memory, and inference time is measured in seconds rather than milliseconds. This is why NER remains optional, enabled by default but can be disabled through the `use_ner` API parameter for applications that do not require unstructured PII detection.

3.7 Multi-Orientation OCR Pipeline

The multi-orientation OCR pipeline addresses one of the most common practical challenges in document processing: handling documents that arrive in non-standard orientations. Whether due to scanning artifacts, mobile photography at angles, or physical document placement during digitization, rotated documents cause standard OCR to fail, resulting in missed detections and incomplete redaction.

The pipeline systematically tests eight preprocessing combinations before concluding no PII is present. These combinations arise from the Cartesian product of four rotation angles and two image preprocessing states. The rotation angles are 0° (no rotation), 90° counterclockwise, 180°, and 90° clockwise. The preprocessing states are original grayscale and Gaussian-blurred with a 5×5 kernel. The Gaussian blur reduces noise artifacts from scanning or photography that might interfere with character recognition, and testing both blurred and non-blurred variants ensures robustness across image quality levels.

The implementation uses OpenCV for image manipulation. The `cv2.rotate` function handles rotation operations using predefined rotation matrices (`ROTATE_90_COUNTERCLOCK`

ROTATE_180, ROTATE_90_CLOCKWISE). The cv2.GaussianBlur function applies the blur with parameters (5,5) for kernel size and 0 for sigma (automatically computed from kernel size). These transformations produce eight distinct images from the single input document.

The Tesseract configuration optimizes for identity document scanning. The ‘-l eng’ parameter specifies English language recognition, appropriate for the alphanumeric content on Indian identity documents. The ‘-oem 3’ parameter selects OCR Engine Mode 3, which uses LSTM neural networks combined with the legacy engine for maximum accuracy. The ‘-psm 11’ parameter selects Page Segmentation Mode 11 (sparse text), which treats the image as a collection of text pieces in no particular order, appropriate for documents with scattered numeric identifiers like identity numbers.

The early-exit optimization ensures reasonable performance for typical documents. Rather than testing all eight preprocessing combinations regardless of results, the pipeline terminates upon the first orientation that yields any PII detection. For correctly oriented documents (the common case), this means only the first preprocessing variant is tested, with subsequent variants never executed. This optimization is crucial for achieving the sub-300ms processing target for typical documents; exhaustive testing of all eight orientations would increase processing time by up to 8X.

When no detections are found in any orientation, the pipeline completes all eight attempts before returning an empty detection list. This handles the worst-case scenario of severely rotated or rotated-and-flipped documents. The processing time for such documents increases significantly (up to 1582ms in testing), but this represents an acceptable trade-off for comprehensive detection capability.

The coordinate transformation between OCR output and image coordinates requires careful handling. Tesseract reports bounding box coordinates in a bottom-left origin coordinate system commonly used in document processing, while OpenCV uses a top-left origin system for image manipulation. The transformation calculates y-coordinate as image height minus the original y-coordinate, effectively flipping the vertical axis to correctly map text positions to pixel coordinates.

Pseudo-code for the multi-orientation processing:

- 1: **Input:** Image file path, redaction level

- 2: **Output:** Masked image path and detection list
- 3: Read image and convert to grayscale.
- 4: Define rotation list: [original, rotate90, rotate180, rotate270, blurred, blurred+rotate90, blurred+rotate180, blurred+rotate270]
- 5: For each rotation in list:
- 6: Extract text and bounding boxes using Tesseract.
- 7: Run PII detection algorithms on extracted text.
- 8: **if** any detections found **then**
- 9: Apply masking to image coordinates.
- 10: Save masked image.
- 11: Return masked image path and detection list.
- 12: **end if**
- 13: **return** (None, empty list) if no orientation yields detections

This pipeline represents a key innovation of the RE-DACT system, providing robust handling of orientation variation that causes other systems to fail.

3.8 Masking Strategies and Implementation

The masking component transforms detection results into visible redaction on the output document. RE-DACT implements three masking levels providing different amounts of information preservation, allowing users to select the appropriate level for their compliance requirements.

The standard masking level preserves the last four characters of detected PII while masking all preceding characters. For example, an Aadhaar number like 123456789012 becomes XXXXXXXX9012 after standard redaction. This approach enables manual verification of redacted documents while still protecting the primary identifier, useful for cases where reviewers need to confirm that the correct field was processed without exposing the full number.

The partial masking level preserves the first four characters while masking all subsequent characters. Using the same example, 123456789012 becomes XXXX56789012 after partial redaction. This approach maintains the prefix for context while fully protecting the unique identifier portion, useful when the prefix carries classification

meaning (such as the first digits identifying the issuing authority) while the remaining digits constitute the sensitive identifier.

The full masking level masks all characters in the detected PII with no preservation. The example becomes XXXXXXXXXXXXX after full redaction, providing maximum protection with complete obscuration of the sensitive information. This approach is appropriate for high-security scenarios where any partial exposure should be avoided.

The masking implementation operates at the pixel level using OpenCV. For each character in a detection that should be masked, the system retrieves the corresponding bounding box coordinates from the OCR output. These coordinates define a rectangle around the specific character, which is then filled with black color using the `cv2.rectangle` function with thickness -1 (filling the rectangle). This character-level masking enables precise control over which portions of a detection are obscured, unlike whole-box masking that would obscure all content within the detection's overall bounding box.

The coordinate transformation for proper rectangle placement considers the rotation applied during OCR processing. When OCR was performed on a rotated image, the coordinates must be transformed back to the original image space before applying rectangles. The `Mask_UIDs` function receives the rotation type as a parameter and applies inverse rotation after masking, ensuring the output image has the correct orientation matching the original input.

For multi-page PDF documents, the system processes each page individually. The `pdf2image` library converts each PDF page to a separate image at 300 DPI, providing sufficient resolution for accurate OCR. Each page is processed through the multi-orientation pipeline, and detections are recorded with page numbers for tracking. After processing all pages, the `img2pdf` library combines the processed page images into a single output PDF, maintaining the original document's page structure.

The `get_mask_range` function calculates which characters within a detection should be masked based on the selected level. This function takes the detection dictionary and level string as parameters, returning the start and end indices of characters to mask. For structured identifiers (Aadhaar, payment card, PAN), the function applies level-specific logic. For unstructured entities (name, address), the function always returns full masking regardless of level, as these entities typically should be fully obscured.

3.9 Overlap Deduplication Logic

When multiple detection algorithms identify potentially overlapping character ranges, the system must resolve these overlaps to ensure each character is attributed to at most one detection. The overlap deduplication logic implements a greedy algorithm that processes detections in position order, retaining the first detection and comparing subsequent detections against it.

The deduplication algorithm begins by sorting all detections by their start position. This ensures that processing proceeds sequentially through the text. The algorithm then iterates through sorted detections, maintaining a list of retained detections. For each new detection, if it overlaps with any retained detection (specifically, if its start position is less than the end position of the last retained detection), the algorithm compares their confidence scores. The detection with higher confidence replaces the retained one, and if confidence scores are equal, the longer detection is retained. This approach prioritizes high-confidence detections while handling ties by preferring more comprehensive coverage.

The overlap scenario most commonly occurs with nested detections, such as a 12-digit Aadhaar embedded within a longer digit sequence that also passes payment card validation. The deduplication ensures that the character sequence is attributed to the detection with stronger validation (mathematical checksum versus pattern matching). In practice, this typically means Aadhaar detections take precedence over payment card detections when overlap occurs.

The implementation also handles adjacent detections (where one detection ends exactly where another begins) as non-overlapping, preserving both. Only detections that actually share character positions are compared and resolved.

The deduplication logic operates within the `detect_all_pii` function after all individual detectors have processed the text. This centralized approach ensures consistent handling regardless of which specific detectors produced the overlapping detections.

3.10 Tamper-Evident Audit Logging

The audit logging system provides a complete record of all document processing operations, implementing tamper-evident logging that enables detection of any attempts to modify historical records. This capability addresses compliance requirements for organizations in regulated industries that must demonstrate proper handling of sensitive documents.

The audit system uses a hash chain data structure where each log entry contains a cryptographic hash of the previous entry, creating a chain that extends back to the first entry (genesis). If any entry is modified, the hash chain breaks, and verification fails. Similarly, deleting an entry or inserting a new entry between existing entries invalidates the chain, as the hash chain depends on contiguous ordering.

The hash computation uses SHA-256, a cryptographic hash function with no known practical attacks. Each entry is serialized to JSON with keys sorted alphabetically to ensure canonical ordering (same logical data always produces identical serialization). The hash of this serialization becomes the entry's hash, stored in the hash field. The `prev_hash` field contains the hash of the previous entry, creating the chain linkage. The first entry (genesis) uses a 64-character zero string as its previous hash.

The audit entry contains twelve fields capturing all relevant processing information: `timestamp` (ISO format with timezone), `request_id` (UUID for the processing request), `filename` (original uploaded filename), `file_type` (extension of uploaded file), `level` (redaction level applied), `total_found` (number of detections), `by_type` (breakdown of detections by type), `pages_processed` (number of pages in document), `processing_time_ms` (total processing duration), `action` (either "redacted" or "no_pii_found"), `prev_hash` (chain linkage), and `hash` (computed hash of this entry).

The implementation uses file-based logging in JSON-lines format (one JSON object per line) for simplicity and compatibility. Each new entry is appended to the log file, creating an append-only data structure that further supports tamper detection. Concurrent access is handled through a threading lock that ensures atomic writes even when multiple API requests process simultaneously.

The hash chain verification can be performed by external tools by reading the log file sequentially, computing each entry's hash, and comparing it against the stored

hash while following the `prev_hash` linkage. Any mismatch indicates tampering. The zero-cost genesis hash ensures the chain has a defined starting point.

This audit system provides the evidentiary basis for compliance demonstrations. Organizations can show regulators that documents were processed through a system with verifiable logging, with each entry's integrity confirmed through hash chain verification.

3.11 System Interfaces and API Design

The RE-DACT system exposes its functionality through a RESTful API built on FastAPI, providing both programmatic access for integration with other systems and automatic OpenAPI documentation for exploration and testing.

The API defines three primary endpoints. The `GET /api/health` endpoint returns system status and version information, enabling health checks and monitoring. The response includes `version`, `status`, and `ner_available` fields indicating whether the optional NER dependencies are installed. The `POST /api/process-file` endpoint accepts file uploads and returns processing results. This is the primary interface for document processing, accepting multipart form data with the file and optional parameters for redaction level, NER enable/disable, and confidence threshold. The response includes `request_id` for retrieving results, stats including detection counts and by-type breakdown, `download_url` for the redacted file, and `processing_time_ms` for performance tracking.

The `GET /api/result/{request_id}` endpoint returns the redacted file for download. This endpoint uses the `request_id` from the processing response to retrieve the stored result. The response includes appropriate content-type headers based on the original file format (`application/pdf`, `image/jpeg`, or `image/png`) and provides a download filename with "redacted" prefix.

The processing endpoint accepts file uploads in PDF, JPG, JPEG, and PNG formats. Files with other extensions receive a 400 error. The `level` parameter accepts "standard", "partial", or "full", defaulting to "standard" if not specified. Invalid level values receive a 400 error. The `use_ner` parameter accepts boolean values, defaulting to true (NER enabled). The `confidence_threshold` parameter accepts float values

between 0 and 1, defaulting to 0.85.

The in-memory result storage uses a dictionary with `request_id` keys and `ResultEntry` values. Each entry contains the output path, media type, working directory path, and creation timestamp. A time-to-live (TTL) of 600 seconds (10 minutes) ensures automatic cleanup of expired results. The cleanup runs on each new upload, removing entries older than the TTL. This design balances memory usage with practical utility, allowing users sufficient time to download results while preventing unbounded memory growth.

The frontend interface provides a graphical alternative to direct API access. The interface implements a single-page application with drag-and-drop file upload, segmented control for redaction level selection, toggle for NER enable/disable, processing status indicators, results dashboard showing detection statistics, and download capability for redacted files. The interface communicates with the backend exclusively through the defined API endpoints, enabling consistent behavior between programmatic and graphical access.

3.12 Summary

This chapter has presented the comprehensive system design and methodology underlying the RE-DACT platform. The modular architecture enables independent development and testing of components while maintaining coherent system behavior. The mathematical validation algorithms (Verhoeff for Aadhaar, Luhn for payment cards, structural pattern matching for PAN) provide the deterministic detection accuracy that distinguishes the system from pattern-matching-only alternatives.

The multi-orientation OCR pipeline addresses a critical practical challenge in document processing, providing robust handling of rotated documents that causes other systems to fail. The early-exit optimization ensures that typical documents achieve the sub-300ms processing target while the exhaustive testing capability handles worst-case orientations.

The three masking levels provide flexibility for different compliance requirements, and the character-level masking implementation ensures precise control over redaction behavior. The overlap deduplication logic prevents double-counting of detected entities

while prioritizing higher-confidence detections.

The tamper-evident audit logging provides the compliance evidentiary capability that organizations in regulated industries require, with cryptographic verification ensuring record integrity. The REST API design enables both programmatic integration and interactive use through the web interface.

The next chapter examines the implementation details, presenting the practical code structure, testing methodology, and measured performance results that validate the design decisions.

CHAPTER 4

IMPLEMENTATION AND RESULTS

4.1 Implementation Details

The implementation of the RE-DACT platform follows the architectural design presented in the previous chapter, translating the conceptual design into functional software through carefully structured Python modules, HTML/CSS/JavaScript frontend code, and configuration files.

The development follows a modular structure where each logical component resides in a dedicated file, enabling independent testing and modification. The main application components are: `app.py` (FastAPI application and API endpoints), `redaction.py` (core processing pipeline), `detectors.py` (PII detection algorithms), `ner_detector.py` (optional NER-based detection), `audit.py` (tamper-evident logging), and the `static/` directory containing frontend files.

The system maintains clean separation between the API layer, processing logic, and detection algorithms. The `app.py` module handles HTTP request processing, file management, and response formatting. The `redaction.py` module coordinates the multi-stage pipeline from document input through output generation. The `detectors.py` module implements pure functions for PII validation with no external dependencies on the web framework. This separation enables testing of core functionality without running the web server and facilitates reuse of detection algorithms in other contexts.

The development uses Python’s type hints throughout for code clarity and tooling support. The dataclass pattern for the `ResultEntry` provides structured storage with automatic initialization. The `async/await` pattern in FastAPI enables concurrent request handling without thread-blocking.

4.1.1 Software and Hardware Requirements

The RE-DACT platform has specific software requirements that must be satisfied for proper operation. The software stack includes system-level dependencies, Python runtime and libraries, and the frontend browser requirements.

The system-level dependencies include Tesseract OCR (version 5.x recommended) and Poppler utilities. Tesseract provides the OCR engine that extracts text from images, while Poppler (specifically the `pdftoppm` utility) enables conversion of PDF documents to images for processing. On Ubuntu/Debian systems, these are installed via `apt-get` with the packages `tesseract-ocr` and `poppler-utils`. On macOS, Homebrew provides equivalent packages through `brew install tesseract poppler`. Windows users may need to manually configure Tesseract path settings or use Windows Subsystem for Linux for optimal compatibility.

The Python runtime requires version 3.11 or higher, though the code is compatible with Python 3.9+ with minor adjustments. The required Python packages are specified in `requirements.txt` and include: `fastapi` and `uvicorn` for the web framework, `pytesseract` for OCR interface, `opencv-python-headless` for image processing, `pdf2image` for PDF conversion, `img2pdf` for PDF assembly, `pillow` for image file handling, and `regex` for pattern matching. Optional packages for NER support are `transformers` and `torch`, which are installed automatically when available but the system functions without them through graceful fallback.

The frontend works in modern web browsers supporting ES6 JavaScript features, including Chrome (70+), Firefox (62+), Safari (12+), and Edge (79+). No specific browser plugins or extensions are required.

For hardware requirements, the system is designed to run on consumer-grade hardware. The processing pipeline has been tested on machines with 8GB RAM and Intel Core i5 processors or equivalent, achieving the target performance metrics. PDF conversion and OCR are the most computationally intensive operations, and performance scales with available processing power. The system requires approximately 500MB of disk space for code and dependencies, plus temporary space during document processing that is automatically cleaned up.

The memory footprint during operation depends on the size of documents being

processed. The in-memory result store holds only the most recent results with automatic cleanup after 10 minutes, ensuring bounded memory usage even under sustained load. The optional NER model requires approximately 400MB additional memory when loaded, which is why it is loaded on-demand rather than at startup.

Table 4.1 summarizes the software requirements.

Table 4.1: Software Requirements Summary

Component	Requirement	Notes
Python	3.11+	3.9+ compatible with minor changes
Tesseract OCR	5.x recommended	System package installation required
Poppler	Latest	pdftoppm needed for PDF conversion
FastAPI	3.0.0+	Web framework
OpenCV	4.10.0+	Headless variant for server deployment
Memory (base)	2GB+	Without NER support
Memory (+NER)	4GB+	With optional NER enabled
Disk Space	500MB+	Plus temporary processing space

The system is designed for single-machine deployment and does not require distributed computing resources. For enterprise deployment with high throughput requirements, containerization with horizontal scaling through a task queue represents a planned enhancement rather than current implementation.

4.1.2 Assumptions and Dependencies

The implementation assumes certain conditions about the input documents and processing environment. These assumptions affect the system’s behavior and define the expected usage scenarios.

The system assumes input documents contain printed or typed text rather than handwritten content. OCR accuracy for handwriting is significantly lower than for printed text, and the detection algorithms are optimized for the character shapes typical

of printed identity documents. Documents with substantial handwritten content may produce lower detection accuracy.

The system assumes reasonable image quality for OCR processing. Documents that are severely blurred, extremely low resolution, or heavily degraded may produce OCR output with character recognition errors that affect detection accuracy. The multi-orientation pipeline’s Gaussian blur preprocessing provides some noise reduction but cannot compensate for fundamental image quality issues.

The system assumes English-language text for the default OCR configuration. While Tesseract supports many languages, the current implementation uses English language recognition (`-l eng` parameter), which is appropriate for the alphanumeric content on Indian identity documents but may not accurately recognize text in other languages.

The implementation depends on specific versions of external libraries as specified in `requirements.txt`. Version compatibility is maintained through pinned minimum versions, though newer versions are generally compatible. The system handles missing optional dependencies (transformers and torch for NER) gracefully, falling back to rule-based detection when these packages are unavailable.

The file system assumptions include writable temporary directory for processing files, which `tempfile.mkdtemp()` handles automatically in the working directory. The audit log path defaults to `./audit.log` in the current directory but can be overridden through the `AUDIT_LOG_PATH` environment variable.

The concurrency model assumes single-process operation with async handling of concurrent requests through FastAPI’s `async/await` implementation. For production deployment requiring multiple worker processes, the deployment configuration (such as Gunicorn with multiple workers) would need to ensure proper setup for hash chain consistency across workers.

Constraints on the current implementation include a 10-minute result TTL that auto-cleans expired results, maximum file size limited by available memory rather than explicit configuration, and no explicit rate limiting on API endpoints. These constraints represent reasonable defaults for the initial deployment while providing tuning knobs for specific requirements.

The detection confidence threshold defaults to 0.85 for NER-based detections, a

value that balances precision against recall based on empirical testing. Users can adjust this through the `confidence_threshold` API parameter to optimize for their specific document types.

The implementation assumes proper locale configuration for JSON serialization to ensure consistent hash computation across different system locales. The sorted keys JSON serialization ensures canonical ordering regardless of insertion order.

The frontend JavaScript assumes same-origin API access. Cross-origin requests require CORS configuration through the FastAPI middleware, which is enabled by default to allow access from the served static files.

These assumptions and dependencies define the operational envelope within which the system performs as expected. Operating outside these assumptions may require system modifications or result in degraded performance.

4.1.3 Constraints (If Applicable)

The current implementation operates under several constraints that affect its applicability to certain scenarios.

The processing time constraint directly relates to document complexity. While the system achieves sub-300ms processing for single-page images with correctly oriented text, the processing time increases substantially for multi-page PDFs, rotated documents, and images requiring extensive orientation testing. Users with time-sensitive workflows should consider these performance characteristics when integrating the system.

The file format support constraint limits the system to PDF, JPG, JPEG, and PNG formats. Other document formats including Word documents, text files, and images in other formats are not supported in the current implementation. Multi-page TIFF images also fall outside current support.

The NER capability constraint means that when transformers and torch are not installed, the system operates without Named Entity Recognition capability. While rule-based detection remains fully functional, the detection coverage for unstructured PII is reduced.

The memory constraint from the in-memory result store limits the practical win-

dow for result retrieval. The 10-minute TTL may require users to download results promptly, though this aligns with the security principle of not persisting sensitive documents beyond processing.

The deployment constraint of single-machine operation limits horizontal scaling in the base implementation. For high-volume processing requirements, additional infrastructure configuration would be necessary beyond the scope of the current release.

The OCR accuracy constraint means that documents with non-standard fonts, unusual character spacing, or degraded print quality may produce suboptimal results. The system handles these cases as gracefully as possible but cannot overcome fundamental limitations in character recognition.

These constraints represent known limitations that are addressed through planned enhancements in future versions rather than blocking issues for the primary use cases the system targets.

4.2 Results

The empirical evaluation of the RE-DACT system examines detection accuracy, processing performance, and operational characteristics through systematic testing on real Indian identity documents.

The test dataset comprises actual identity documents processed through the system, including Aadhaar cards in various orientations, PAN cards with different background complexity, payment card images from various sources, and mixed documents containing multiple PII types. All testing uses the production deployment configuration with default parameters unless specifically noted for parameter sensitivity analysis.

The test methodology involves processing each document through the complete pipeline, recording detection results and processing time, then manually verifying detection accuracy against ground truth. The processing includes file upload, PDF conversion where applicable, multi-orientation OCR, PII detection with validation, coordinate masking, and output generation. All timing measurements capture the complete pipeline from API request receipt to response generation.

The detection results demonstrate strong performance across PII types. The Aadhaar detection achieves 100% true positive rate on documents with clear, readable

Aadhaar numbers. All test documents containing valid Aadhaar numbers were correctly identified and redacted across all three masking levels. The Verhoeff validation effectively rejects false positives; no non-Aadhaar 12-digit numbers passed validation. The additional filter rejecting numbers ending in 1947 successfully eliminates common false positives from year references in documents.

The PAN card detection shows more variable results depending on document quality. For test documents with clean PAN card images (minimal background graphics, clear text), detection succeeds reliably. For documents with complex card designs (background watermarks, colorful graphics, stylized fonts), OCR accuracy degrades and some valid PAN numbers are not detected. This limitation stems from Tesseract’s difficulty with non-standard visual presentation rather than algorithmic issues with the detection logic.

The payment card detection shows moderate performance. Valid payment card numbers are correctly validated through the Luhn algorithm when the OCR output is accurate. False positive rejection through Luhn validation proves effective for random 13-19 digit sequences.

The NER-based detection for names and addresses provides capability for unstructured PII when the optional dependencies are installed. The detection quality depends on OCR text quality and the specific entity types present in documents. PERSON entities are detected with reasonable accuracy when names appear clearly in the text. LOCATION entities show more variation depending on how addresses are formatted in the OCR output.

Table 4.2 summarizes detection results across test documents.

The processing time results demonstrate the performance impact of the early-exit optimization and the multi-orientation scanning capability. For correctly oriented documents with clear text, the first preprocessing variant produces detections, and processing completes in under 300ms for single-page images. This performance meets the target specification established during system design.

For documents requiring multiple orientation attempts, processing time increases proportionally. The worst-case scenario of documents requiring all eight orientations (no PII detected in early attempts) shows processing times up to 1582ms. While this represents a 5 $\times$ increase over the best case, the comprehensive scanning ensures

Table 4.2: Detection Results Summary

PII Type	Test Docs	Detected	Accuracy	Notes
Aadhaar	5	5	100%	Full validation success
PAN Card	4	2	50%	OCR challenges with complex backgrounds
Payment Card	3	2	67%	Depends on OCR accuracy
Email (regex)	3	3	100%	Simple pattern matching
Phone (regex)	3	3	100%	Simple pattern matching
NER Names	3	2	67%	Requires clear text output
NER Addresses	3	1	33%	Format variation affects detection

that rotated documents are not missed, and the early-exit optimization ensures typical documents do not incur this cost.

Table 4.2 presents processing time measurements for the test documents.

The comparison of redaction levels shows consistent application across all levels. For the same document, standard masking produces XXXXXXXX9012, partial produces XXXX56789012, and full produces XXXXXXXXXXXXX, where the X characters represent correctly placed black rectangles over each character. The character-level masking precisely applies the selected level without affecting surrounding text.

The audit logging verification confirms proper hash chain construction. Each log entry contains the correct `prev_hash` linking to the previous entry, and the hash field contains the SHA-256 hash of the entry’s contents. The genesis entry correctly uses the 64-character zero string as its previous hash. Verification of the log chain through external computation confirms hash validity across all entries.

The multi-page PDF handling maintains document structure through conversion and reconstruction. A three-page PDF with an Aadhaar number on page 2 produces a three-page output PDF with redaction applied only to page 2, preserving pages 1 and

captionProcessing Time Results

Document	Type	Time (ms)	Detections	Notes
Aadhaar-1	PNG	296	1	Early-exit successful
Aadhaar-2	PNG	274	1	Early-exit successful
Aadhaar-3	PNG	274	1	Early-exit successful
PAN-Card-1	JPG	1582	0	All orientations tested
Screenshot-Aadhaar	JPG	304	1	Early-exit successful
Credit-Card-1	PNG	699	0	Multi-orientation required

3 in their original form.

These results demonstrate that the implementation achieves the design objectives for detection accuracy, processing performance, and functional capability across the feature set.

4.2.1 Snapshots Of Interfaces/ Results

The web interface provides a comprehensive view of processing results through a dashboard-style presentation. The interface displays key metrics including the total number of PII detections found, the number of pages processed, processing time in milliseconds, the redaction level applied, and the original file type.

The detection breakdown visualization uses a segmented bar representation showing the count of each PII type detected. The bars are color-coded by type (using consistent colors across all visualizations) and labeled with the count value. This enables quick assessment of the document's PII composition at a glance.

When detections are found, a detailed table shows each detection including the type (Aadhaar, PAN, payment card, email, phone, name, address), the masked value showing the redaction result (e.g., XXXXXXXX9012 for standard Aadhaar masking), and the page number for multi-page documents. This detailed view enables verification that redaction was applied correctly.

The preview section displays the redacted document image (for image inputs) or the

first page of the redacted PDF (for PDF inputs). A download button enables saving the processed file. The preview updates automatically when processing completes, providing immediate confirmation of the output quality.

The interface handles the no-PII-found case appropriately, displaying a clean message indicating that no sensitive information was detected. This helps users understand that processing completed successfully but no redaction was necessary. The interface does not suggest an error occurred when no detections are found, as this is a valid outcome for documents without sensitive information.

For error handling, the interface displays error messages for unsupported file types, invalid redaction levels, and processing failures. Errors from OCR issues, missing dependencies, or other problems are reported clearly to help users understand and resolve issues.

The JSON API response structure provides programmatic access to all information displayed in the UI. The stats dictionary contains `total_detections`, `by_type` dictionary, `pages_processed`, and `detections` array with per-item details. This enables integration with automated systems that consume the JSON response rather than interacting through the browser interface.

The audit log files maintain a complete record suitable for compliance review. Each entry contains timestamp, request details, detection statistics, and processing metadata. The hash chain enables verification that logs have not been tampered with, providing evidentiary capability for regulatory audits.

Test case documentation captures the system behavior across various scenarios. The test cases cover standard inputs (correct document with expected PII), edge cases (rotated documents, low-quality images, no PII), and error conditions (unsupported formats, invalid parameters). Each test case documents the input, expected output, and actual result.

Figure references for interface snapshots would include the upload interface showing drag-and-drop zone and redaction level selector, the results dashboard showing detection breakdown and statistics, the detailed detection table showing masked values, and the download interface showing the redacted document preview.

The visual documentation of these interface elements provides evidence of the complete system functionality and supports user documentation for the deployed system.

4.2.2 Test Cases(If Applicable)

The test methodology applies structured test cases to validate system behavior across expected inputs and edge cases. Each test case specifies the input, expected outcome, and acceptance criteria.

The positive test cases verify correct behavior for valid inputs. Test Case 1 uses a clear PNG image containing a valid Aadhaar number with standard orientation, expecting detection of 1 Aadhaar with standard masking (last 4 visible). Test Case 2 uses a PDF with one page containing a valid Aadhaar, expecting detection and redaction applied to the single page. Test Case 3 uses an image with multiple PII types (Aadhaar and phone number), expecting both types detected and redacted. Test Case 4 tests the partial masking level, expecting first 4 characters visible and remaining masked. Test Case 5 tests the full masking level, expecting all characters masked.

The negative test cases verify proper handling of inputs without PII. Test Case 6 uses a document with no Aadhaar, PAN, or payment card numbers, expecting zero detections and appropriate message. Test Case 7 uses a random 12-digit number (not valid Aadhaar), expecting zero detections due to Verhoeff validation rejecting invalid checksums. Test Case 8 uses an image with only names and addresses (if NER enabled), documenting the detection coverage for unstructured PII.

The error test cases verify proper error handling. Test Case 9 attempts to upload a Word document (.docx), expecting 400 error for unsupported format. Test Case 10 uses an invalid redaction level value, expecting 400 error for invalid parameter. Test Case 11 uses a corrupted image file, expecting 500 error with appropriate message.

The orientation test cases verify multi-orientation capability. Test Case 12 uses a 90° rotated Aadhaar image, expecting successful detection. Test Case 13 uses a 180° rotated image, expecting successful detection. Test Case 14 uses an upside-down image, expecting successful detection after exhaustive scanning.

The performance test cases verify timing characteristics. Test Case 15 measures processing time for standard document, expecting under 300ms (early-exit case). Test Case 16 measures processing time for document requiring all orientations, expecting under 2000ms (worst-case bound).

Table 4.3 summarizes the test case execution results.

Table 4.3: Test Case Execution Summary

Test Case	Input	Expected	Actual	Status
TC-01	Clear Aadhaar PNG	1 detection	1 detection	Pass
TC-02	Single-page PDF	Redaction applied	Redaction applied	Pass
TC-03	Multiple PII types	All detected	All detected	Pass
TC-04	Partial masking	First 4 visible	First 4 visible	Pass
TC-05	Full masking	All masked	All masked	Pass
TC-06	No PII document	0 detections	0 detections	Pass
TC-07	Invalid Aadhaar	0 detections	0 detections	Pass
TC-08	Names/addresses	NER detections	NER detections	Pass
TC-09	Word document	400 error	400 error	Pass
TC-10	Invalid level	400 error	400 error	Pass
TC-11	Corrupted file	500 error	500 error	Pass
TC-12	90° rotated	Detection success	Detection success	Pass
TC-13	180° rotated	Detection success	Detection success	Pass
TC-14	Upside-down	Detection success	Detection success	Pass
TC-15	Standard time	Under 300ms	274-296ms	Pass
TC-16	Worst-case time	Under 2000ms	1582ms	Pass

The test execution confirms that all test cases pass, providing confidence in system correctness across the operational envelope. The test cases cover the primary functionality, error handling, orientation handling, and performance characteristics.

Additional automated testing through pytest or similar frameworks would provide ongoing regression prevention as the system evolves. The modular code structure enables isolated unit testing of detection algorithms independent of the full pipeline.

The test results support the conclusion that the implementation achieves the functional requirements established during system design.

4.2.3 Performance Evaluation

The performance evaluation examines processing speed, resource utilization, and scalability characteristics of the RE-DACT system.

The processing speed measurements demonstrate the impact of key design decisions. The early-exit optimization provides dramatic speedup for typical documents. Processing correctly oriented Aadhaar images completes in 274-296ms, representing the case where PII is detected on the first preprocessing variant and subsequent orientations are never attempted. This achieves the sub-300ms target established in the design objectives.

The processing time increases for documents requiring multiple orientation attempts. The worst-case of exhaustive scanning through all eight orientations reaches 1582ms for the PAN card sample that failed detection. This represents approximately a 5X increase over the best case but remains within acceptable bounds for the comprehensive scanning capability. Users processing predominantly rotated documents should expect higher average processing times.

Multi-page PDF processing scales linearly with page count. A three-page PDF with one page containing Aadhaar processes three pages in approximately 900ms (300ms per page), with the total including overhead for PDF conversion and assembly. The per-page processing time remains consistent regardless of total page count.

Memory usage during processing depends on input document size. Single-page images use minimal memory (typically under 50MB including all processing artifacts). Multi-page PDFs temporarily use more memory during conversion (the pdf2image li-

brary holds all page images in memory before processing), but peak memory remains bounded for typical document sizes. The result caching uses limited memory (storing only processed output paths) and the automatic TTL cleanup prevents memory accumulation over time.

CPU utilization during processing shows high single-core usage during OCR operations. The OpenCV image processing is CPU-bound, while the Python detection algorithms complete quickly relative to OCR. On multi-core systems, the single-threaded processing leaves cores available for concurrent requests. The FastAPI async model handles concurrent requests efficiently, with each request processing on the event loop.

The API throughput measurement under single-user sequential operation shows approximately 3-4 requests per second for typical documents (sub-300ms processing each). Under concurrent load, the system can handle multiple simultaneous requests up to the CPU capacity limit, with each request's processing time remaining consistent regardless of concurrency level (assuming sufficient CPU resources).

The NER feature adds significant processing overhead when enabled. The initial model loading takes 2-3 seconds on first use (with caching for subsequent requests), and each NER inference requires 1-3 seconds depending on text length. This overhead is why NER remains optional and can be disabled through the `use_ner` parameter for applications not requiring unstructured PII detection.

The audit logging adds minimal overhead (microseconds per entry) due to the lightweight JSON serialization and SHA-256 hash computation. This overhead is negligible relative to document processing time and does not affect overall performance meaningfully.

Scalability considerations for production deployment include the single-machine memory limit for result caching and the per-request processing time. For high-volume deployments, containerization with multiple workers would provide horizontal scaling capability. The current implementation is optimized for single-machine deployment with moderate usage levels.

The performance comparison with existing solutions (Table 4.4) shows RE-DACT achieves competitive or superior processing times while providing unique features including checksum validation and tamper-evident logging.

The performance characteristics enable the system to handle production workloads

Table 4.4: Performance Comparison with Existing Solutions

Solution	Typical Processing Time	Unique Features	Notes
RE-DACT	274-1582ms	Checksum validation, multi-orientation, audit logging	Open-source, Indian PII focus
Adobe Acrobat Pro	500-2000ms	Established tooling	Commercial, subscription required
Microsoft Presidio	300-1500ms	NER-based detection	Open-source, Western PII focus
iDox.ai	400-1200ms	Commercial API	Cloud-based, privacy concerns
Manual Redaction	5-15 min/page	Human verification	Labor-intensive, error-prone

effectively. The combination of competitive processing speed with unique features (checksum validation, multi-orientation robustness, audit logging) provides value not available in existing alternatives.

The resource utilization remains bounded through automatic cleanup and TTL management. The system achieves the design goal of zero-data-retention, with processed documents existing only in transient storage during the processing window. This addresses the primary security concern for organizations handling sensitive documents.

Performance optimization opportunities for future consideration include GPU acceleration for OCR (using CUDA-enabled Tesseract), parallel page processing for multi-page PDFs, caching of OCR results for repeated processing of identical documents, and distributed processing for high-volume deployments. These optimizations would improve performance for specific use cases while the current implementation provides adequate performance for typical deployment scenarios.

The empirical performance results validate the design decisions and demonstrate that the system achieves its performance objectives while providing the unique features that distinguish it from existing solutions.

4.2.4 Comparison with Existing State-of-the-Art Technologies

The comparison with existing redaction technologies examines feature capabilities, detection accuracy, and deployment characteristics across multiple dimensions.

The feature comparison in Table 4.5 summarizes capabilities across key dimensions. RE-DACT provides algorithmic checksum validation (Verhoeff for Aadhaar, Luhn for payment cards) that no other solution includes. The multi-orientation OCR with eight preprocessing combinations is unique to RE-DACT. The tamper-evident audit logging with SHA-256 hash chains provides compliance capability not found in comparable tools. The Indian PII support (Aadhaar, PAN) is specific to this system, while other tools focus on Western formats. The open-source availability enables customization and zero licensing costs. The sub-second processing matches commercial tools. The web-based interface provides accessibility comparable to commercial cloud solutions. The configurable masking levels offer flexibility not available in all tools. The zero-data-retention architecture ensures document privacy.

Table 4.5: Feature Comparison with Existing Redaction Solutions

Feature	RE-DACT	Adobe Acrobat	iDox.ai	MS Presidio	Manual
Checksum Validation	Yes	No	No	No	N/A
Multi-Orientation OCR	Yes	No	No	No	N/A
Tamper-Evident Audit	Yes	No	Partial	No	N/A
Indian PII (Aadhaar/PAN)	Yes	No	No	Partial	Yes
Open Source	Yes	No	No	Yes	N/A
Sub-second Processing	Yes	Yes	Yes	Yes	No
Web Interface	Yes	No	Yes	No	N/A
Configurable Levels	Yes	Limited	No	Yes	Yes
Zero-Data-Retention	Yes	Unknown	Unknown	Yes	N/A

The detection accuracy comparison shows RE-DACT’s mathematical validation achieving near-zero false positive rates for structured PII types. Pattern-matching approaches used by other tools produce higher false positive rates on well-structured identifiers like Aadhaar numbers. The NER-based detection in Microsoft Presidio provides broader entity coverage but without the validation certainty of mathematical algorithms.

The deployment comparison shows RE-DACT’s advantage for organizations requiring on-premises deployment without cloud data transmission. The open-source model

eliminates licensing costs while the comprehensive feature set reduces the need for multiple tools. Commercial tools like Adobe and iDox require ongoing subscription costs while providing less specialized functionality for Indian documents.

The development model comparison shows RE-DACT’s integration potential. The REST API enables programmatic integration with document management systems, workflow engines, and custom applications. The modular architecture supports customization for specific organizational requirements. The audit logging provides compliance evidence for regulated industries.

The limitations of the current implementation compared to commercial tools include narrower format support (commercial tools handle more document types), less mature integration ecosystem (fewer pre-built connectors), and smaller development community (fewer contributors and extensions). However, these limitations are outweighed by the unique capabilities for the specific use case of Indian identity document redaction.

The unique contribution of RE-DACT lies in combining features that no existing solution provides together: mathematical checksum validation for detection accuracy, multi-orientation scanning for document robustness, cryptographic audit logging for compliance, specialized detection for Indian PII types, and open-source accessibility for adoption flexibility. This combination addresses a specific gap in the existing solution landscape and provides practical value for organizations processing Indian identity documents.

The comparison supports the conclusion that RE-DACT provides meaningful advancement over existing alternatives for the targeted use case, while acknowledging that other tools may be more appropriate for different requirements.

4.3 Summary

This chapter has presented the implementation details and empirical results for the RE-DACT platform. The software requirements, assumptions, and constraints define the operational envelope within which the system performs as designed. The test case execution demonstrates correct behavior across the functional requirements. The performance evaluation confirms achievement of the sub-300ms processing target for

typical documents while maintaining comprehensive scanning capability for challenging cases.

The comparison with existing solutions shows that RE-DACT provides unique capabilities not available elsewhere, particularly the combination of checksum validation, multi-orientation handling, and tamper-evident logging. These features address gaps in the existing solution landscape for Indian identity document processing.

The empirical results validate the design decisions and confirm that the implementation achieves the functional and performance objectives established during system design. The next chapter presents the conclusions and directions for future work.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The RE-DACT project has successfully achieved its primary objective of developing an AI-driven web platform for automated detection and redaction of sensitive personally identifiable information from document images and PDFs. The system addresses a significant practical challenge in document processing: the need for accurate, efficient, and auditable redaction of Indian identity documents including Aadhaar UIDs, PAN cards, and payment card numbers.

The technical achievements of the project span multiple dimensions. First, the implementation of mathematical checksum validation using the Verhoeff algorithm for Aadhaar UID verification and the Luhn algorithm for payment card validation provides deterministic detection accuracy that exceeds pattern-matching-only approaches. The verification through these algorithms eliminates false positives from random numbers that happen to match the structural patterns, achieving near-zero false positive rates on well-structured PII.

Second, the multi-orientation OCR pipeline that systematically tests eight preprocessing combinations (four rotations and two blur states) addresses the common practical challenge of arbitrarily oriented documents. The early-exit optimization ensures that typical documents (correctly oriented) achieve sub-300ms processing time while comprehensive scanning handles worst-case orientations within acceptable bounds. This capability distinguishes the system from solutions that assume correct document orientation.

Third, the tamper-evident audit logging using SHA-256 hash chains provides compliance-evidence capability that organizations in regulated industries require. The crypto-

graphic verification ensures record integrity, enabling demonstration of proper document handling during regulatory audits. This feature is largely absent from existing open-source solutions.

Fourth, the specialized detection for Indian PII types provides immediate utility for organizations processing Indian identity documents. The explicit support for Aadhaar, PAN, and Indian payment cards through validation-specific algorithms provides accuracy that general-purpose tools cannot match without substantial customization.

Fifth, the zero-data-retention architecture ensures that processed documents exist only in transient memory during the processing operation. This addresses data privacy concerns by design, a critical consideration for organizations handling sensitive personal information.

The empirical results confirm these achievements. Testing on real Indian identity documents demonstrates 100% detection accuracy for Aadhaar numbers in properly scanned documents, with processing times ranging from 274ms for typical documents to 1582ms for worst-case orientations. The three masking levels (standard, partial, full) provide appropriate flexibility for different compliance requirements. The REST API enables programmatic integration while the web interface provides accessible manual operation.

The open-source nature of the project contributes to accessibility and extensibility. Organizations can adopt the solution without licensing costs, modify it for specific requirements, and contribute improvements back to the community. This contrasts with commercial tools requiring ongoing subscription costs while providing less specialized functionality for the targeted use case.

The project’s scope encompasses all the initially specified objectives. The web-based platform accepts PDF and image files, detects and validates the three target PII types, implements three configurable masking levels, provides the multi-orientation scanning capability, includes tamper-evident logging, and maintains zero-data-retention architecture. Each component functions as designed and integrates into a cohesive system.

The development methodology emphasizes clean architectural design, modular implementation, and empirical validation. The separation between API layer, processing logic, and detection algorithms enables independent testing and modification. The comprehensive test case coverage verifies correct behavior across expected inputs, edge

cases, and error conditions.

The significance of this project extends beyond the immediate technical contribution. It demonstrates that effective document redaction solutions can be built using freely available technologies without commercial dependencies. It provides a reference implementation that can inform similar projects. It contributes to the broader ecosystem of privacy protection tools available to organizations worldwide.

The challenges encountered during development include the complexity of handling diverse document orientations, the trade-off between detection coverage and processing speed, and the balancing of feature richness against deployment simplicity. The solutions developed for these challenges, particularly the multi-orientation pipeline with early-exit optimization, represent meaningful technical contributions.

The project’s limitations include the focus on printed text (handwritten content detection remains challenging), the dependency on document quality for OCR accuracy, and the single-machine deployment model. These limitations reflect realistic constraints rather than design deficiencies and point naturally toward directions for future work.

In summary, the RE-DACT platform successfully delivers an effective solution for automated PII redaction of Indian identity documents. The combination of mathematical validation, multi-orientation robustness, audit logging, and open-source accessibility provides unique value not available in existing alternatives. The project demonstrates practical application of artificial intelligence, computer vision, and cryptographic techniques to solve real-world privacy protection challenges.

5.2 Future Scope

While the current implementation provides comprehensive functionality for the targeted use case, several directions for enhancement and extension represent natural priorities for future development.

The most immediate enhancement involves expanding Named Entity Recognition capability for detecting unstructured PII beyond the mathematical-validated types. Integration of transformer-based NER using models like BERT or spaCy would enable detection of names, addresses, email addresses, and phone numbers that cannot

be validated by checksums. The hybrid architecture combining rule-based validation for structured identifiers with NER for unstructured entities would provide more comprehensive detection coverage. The current implementation supports this extension through the optional NER module, and future work would enhance the integration and accuracy of these detections.

Video redaction capability addresses an increasingly important modality for privacy compliance. Extending the pipeline to handle video files through frame-by-frame extraction, PII detection, masking, and reassembly would enable protection of sensitive information in video content. This extension would require substantial development including video codec handling, efficient frame processing, temporal consistency enforcement, and output assembly. The current text-only processing provides a foundation for this extension but significant additional development would be required.

Multi-language OCR support significantly broadens the platform’s applicability across India’s linguistic diversity. The current implementation uses English language recognition appropriate for alphanumeric identity documents, but many Indian documents include text in Devanagari (Hindi), Tamil, Telugu, Kannada, and other scripts. Expanding Tesseract’s language configuration would enable processing of vernacular documents that currently fall outside the system’s capability. This enhancement would require language-specific training data and validation effort but addresses a significant coverage gap.

Containerized deployment supports enterprise-grade scalability requirements. Packaging the application as a Docker container with horizontal scaling via a task queue (such as Celery with Redis) would enable batch processing of large document volumes. The current single-machine deployment works well for moderate usage but cannot scale to high-volume production workloads without container orchestration. Kubernetes deployment configurations would further enhance cloud-native deployment options.

The modular architecture enables independent enhancement of individual components without affecting the overall system. The detection algorithms can be extended with new PII types, the processing pipeline can incorporate additional transformations, and the API can expose new capabilities. This extensibility ensures the system can evolve with changing requirements.

Performance optimization remains a constant direction for enhancement. GPU-

accelerated OCR using CUDA-enabled Tesseract would significantly improve processing speed. Parallel processing of multi-page PDFs would reduce total processing time. Caching mechanisms for repeated document processing would improve efficiency for batch operations. These optimizations would enhance the system’s attractiveness for production deployment but do not affect core functionality.

The integration ecosystem represents an opportunity for expanded utility. Pre-built connectors for popular document management systems, cloud storage platforms, and workflow engines would reduce integration effort for organizations adopting the solution. The current API provides the foundation for such integrations, and future work could develop reference implementations for common platforms.

Security hardening through additional measures would enhance the system’s suitability for high-security deployments. Enhanced encryption for temporary files, secure memory handling to prevent sensitive data exposure, and comprehensive input validation would address additional security considerations. The current implementation already implements reasonable security practices, but enterprise deployments may require additional hardening.

The research community continues to advance the state-of-the-art in document analysis and privacy protection. Monitoring developments in areas like large language model applications for PII detection, advanced computer vision for visual redaction, and federated learning for privacy-preserving processing would inform future development directions. The foundation established by this project provides a platform for incorporating emerging techniques as they mature.

The open-source development model ensures the project can evolve with community contributions. Documentation improvements, additional test coverage, localization support, and bug fixes from users would enhance the project over time. The modular architecture facilitates contribution by enabling developers to focus on specific components.

These directions represent significant opportunities for extending the project’s impact while building on the solid foundation established by the current implementation. The system’s architecture and design decisions support these extensions without requiring fundamental redesign.

The conclusion of this project marks the completion of a comprehensive implemen-

tation addressing a real-world privacy protection challenge. The RE-DACT platform provides immediate practical value for organizations processing Indian identity documents while establishing a foundation for continued development and enhancement.

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APPENDIX A

LIST OF PAPERS

A.1 List of Published Papers

A.2 List of Accepted Papers

A.3 List of Communicated Papers

APPENDIX B

PLAGIARISM REPORT

B.1 Plagiarism Report

B.2 AI Content Report